

TextMamba3D — Project Brief

Background and Motivation: 3D brain tumor segmentation is a critical pre-step for radiotherapy planning and surgical navigation, with BraTS (4-channel MRI: T1 / T1ce / T2 / FLAIR) as the field's standard benchmark. Recent literature (e.g., TextBraTS, MICCAI 2024) has reported that feeding the paired clinical radiology report as an additional input lets SwinUNETR-class models gain about +1.5% Dice on BraTS2020, suggesting an apparently clean promise — text behaves as a "free prior" that should universally help. **However, this promise has never been rigorously validated under a unified framework:** when the backbone, the pretraining strategy, and the fusion mechanism are systematically varied, does the text gain still hold?

Research Question: Can a clinical radiology report, used as an auxiliary modality, genuinely improve 3D medical image segmentation?

Project Carrier and Contribution Definition: To answer this question under tightly controlled variables, we built TextMamba3D — a unified text-guided segmentation framework built entirely on the Mamba (SSM) architecture. We chose Mamba because its $O(n)$ complexity over 3D voxels is better suited to the 1M+ voxels of medical volumes than Transformer-style methods ($O(n^2)$), and because, to our knowledge, no prior work has unified Mamba $\times$ text guidance $\times$ 3D medical segmentation. The framework has ~60M trainable parameters, is trained on A100 / H100 GPUs, and uses BraTS2020 + TextBraTS (369 paired expert text descriptions) as the primary evaluation testbed, with BraTS2021 (1251 cases) as the pretraining source. **The core contribution is jointly defined by two hard criteria:** (1) **absolute Mean Dice strictly higher than TextBraTS SOTA (0.853), with a 1–2 percentage-point margin (i.e., ≥ 0.863)**, demonstrating that the Mamba architecture can clearly surpass SwinUNETR-class Transformer methods on 3D brain tumor segmentation; (2) **text delta significantly positive, and not lower than the +0.55% level we have already achieved on the from-scratch route**, demonstrating that text remains meaningful on top of this strong backbone. Both conditions **must hold simultaneously** — there is an intrinsic trade-off at play here: **the stronger a backbone's ability to learn image features, the more it suppresses the text contribution**, so the text delta is not required to match TextBraTS's +1.5% (that window only exists for weak backbones), but we must prove that text has not been entirely absorbed by the strong backbone.

Core Finding and the Current Bottleneck: After 10+ experimental versions, **none of our versions yet satisfies both hard criteria at the same time**. Concretely, the two main routes fail on different axes, exposing a clean zero-sum pattern:

- **From-scratch route (V5.0 / V5+V6+V7 ensemble):** text delta = +0.55%, exactly meeting the contribution floor and confirming that text is in principle effective on a Mamba backbone. But absolute Mean Dice is only **0.8479 (V5.0 single model) / 0.8511 (3-model ensemble)** — **below TextBraTS SOTA (0.853)**, and far short of the "1–2 percentage points above SOTA" target. **Criterion (1) is not met.**

- **Same-domain pretraining route (V8.0):** absolute Mean Dice jumps to **0.8753** (ET=0.8116, TC=0.8977, WT=0.9167), about 2 percentage points above SOTA — **criterion (1) is met**, which at least proves that "reaching Mamba-based absolute performance 1–2pp above SOTA on brain tumor segmentation" is in fact achievable. But the price is that the **text delta collapses precisely to 0.00%** — the text contribution is 100% absorbed by the strong backbone, and **criterion (2) is completely dead**. This is merely "a Mamba segmenter that is strong without needing text," which is by itself a counter-example to the thesis.

To rule out "maybe we just picked the wrong fusion method," we ran a systematic ablation on top of the V8.0 backbone, sequentially replacing or stacking 9 different text-utilization mechanisms (SeqCA cross-attention, MambaConcatFusion native scan fusion, 10-epoch frozen warmup, 3-stage fully frozen training, vision modality dropout, InfoNCE alignment loss, feature noise injection, decoder-side FiLM, and R-Super-style "text as supervision"). **All nine yielded text delta = 0.00%.** This points to an upstream issue: same-domain pretraining has caused the backbone to absorb, from images alone, all the information that text could possibly provide — tumor location, morphology, enhancement pattern — leaving text structurally redundant in the information-theoretic sense. This also explains why TextBraTS achieves a +1.5% text gain: its SwinUNETR is pretrained on ImageNet (cross-domain), knows nothing about brain tumors, and therefore leaves a "knowledge gap" for text to fill.

Current Experiment (V10.2): We are pretraining the backbone on BraTS2021 with self-supervised Masked Image Modeling — letting it learn only anatomical structure, with no exposure to tumor segmentation labels — to deliberately create a "knows-anatomy, does-not-know-tumors" weak backbone, so that text can fill this tumor knowledge gap during downstream fine-tuning. This is our current and final attempt to break the "strong backbone × effective text" zero-sum.

Next Steps: V10.2's success criterion strictly mirrors the two hard criteria of the core contribution — **absolute Mean Dice ≥ 0.863 (at least 1 percentage point above TextBraTS SOTA)** and **text delta $\geq +0.55\%$ (no lower than what the from-scratch route has already achieved)** — both must hold simultaneously. V8.0 has already proven that the absolute-performance axis is reachable; V5.x has already proven that the text-delta axis is reachable on a weak backbone; what V10.2 must do is **make the two coexist on a single model for the first time**. If V10.2 succeeds, the "strong backbone × effective text" zero-sum is broken, TextMamba3D becomes the *first text-guided Mamba framework strictly beating TextBraTS SOTA on both axes*, and we move into paper writing. If V10.2 fails, we will need to reframe the project's claim boundary — from "proving text is effective" to "characterizing when text becomes ineffective."